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Sediment carbon fate in phreatic karst (Part 1): Conceptual model development

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ABSTRACT

Recent research has paid increased attention to quantifying the fate of carbon pools within fluvial networks, but few, if any, studies consider the fate of sediment organic carbon in fluviokarst systems despite that karst landscapes cover 12% of the earth's land surface. The authors develop a conceptual model of sediment carbon fate in karst terrain with specific emphasis upon phreatic karst conduits, i.e., those located below the groundwater table that have the potential to trap surface-derived sediment and turnover carbon. To assist with their conceptual model development, the authors study a phreatic system and apply a mixture of methods traditional and novel to karst studies, including electrical resistivity imaging, well drilling, instantaneous velocimetry, dye tracing, stage recording, discrete and continuous sediment and water quality sampling, and elemental and stable carbon isotope fingerprinting.

Results show that the sediment transport carrying capacity of the phreatic karst water is orders of magnitude less than surface streams during storm-activated periods promoting deposition of fine sediments in the phreatic karst. However, the sediment transport carrying capacity is sustained long after the hydrologic event has ended leading to sediment resuspension and prolonged transport. The surficial fine grained laminae occurs in the subsurface karst system; but unlike surface streams, the light-limited conditions of the subsurface karst promotes constant heterotrophy leading to carbon turnover. The coupling of the hydrological processes leads to a conceptual model that frames phreatic karst as a biologically active conveyor of sediment carbon that recharges degraded organic carbon back to surface streams. For example, fluvial sediment is estimated to lose 30% of its organic carbon by mass during a one year temporary residence within the phreatic karst. It is recommended that scientists consider karst pathways when attempting to estimate organic matter stocks and carbon transformation in fluvial networks.
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1. Introduction

Fluvial networks are recognized to not only act as conveyors of sediment organic carbon to the ocean, but also to serve as ecosystems that can actively turnover carbon (Battin et al., 2008). Sediment carbon enters the fluvial system via multiple routes which include overland runoff, subsurface flow, mass wasting, and abscission as well as from autochthonous growth within the fluvial system (Ford and Fox, 2014; Hotchkiss and Hall, 2015). It is now recognized that sediment carbon is an important energy source for decomposers and that microbial oxidation results in the production of carbon dioxide and increasingly degraded terrestrially-derived carbon longitudinally in a fluvial system (Swift et al., 1979; Moore et al., 2004). However, the degradation state of sediment carbon and its downstream fate remain highly uncertain with open questions regarding the spatial variability of turnover, temporary burial, and removal of sediment carbon from active carbon cycles (Cole et al., 2007). In this context, one area that has not been well investigated is sediment carbon fate in fluvial systems that drain karst landscapes.

Karst landscapes are typified as solutionally dissolved landscapes that are dominated by secondary and tertiary porosity features (e.g., macropores, fractures, and conduits) that produce low-resistance pathways for water transport (Thrailkill, 1974; Smart and Hobbs, 1986; Pronk et al., 2009b). When coupled to surface streams of the fluvial network, mature karst topography is well-recognized to include subterranean fluid pathways that act as turbulent conduits conveying fluid from surface sinks termed swallets to sources called springs (White, 2002). Karst watersheds often

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carry high loads of sediment brought in by sinking streams and other karst features (Drysdale et al., 2001). In this manner, karst topography provides subsurface pathways for water, sediment, and carbon transport whereby both terrestrially- and aquatically-derived sediment carbon can be temporarily sequestered and transformed only to resurface further downstream. It is highly reasonable that temporarily stored sediment carbon is oxidized and results in a net production of CO_2 given that bacteria and other microbes within epilithic biofilms in subsurface karst utilize particulate and dissolved organic carbon as an energy source (Chapelle, 2001; Danovaro et al., 2001; Simon et al., 2003, 2007; Goldscheider et al., 2006; Humphreys, 2006). Accounting for the spatiotemporal distribution and variability of organic matter inputs, turnover, and fluxes has been identified as one of the greatest challenges in estimating sediment carbon fate in karst (Simon et al., 2007; Pronk et al., 2009a). Thus, the motivation of this paper is towards elucidating the role of hydrologic processes impacting sediment carbon in fluviokarst landscapes and working towards a conceptual model of sediment carbon fate within fluviokarst systems.

A precursor to a conceptual model of sediment carbon impacted by karst is the non-trivial task of estimating the morphology of karst systems, hydraulics of karst water conveyance, and physics of subsurface sediment transport within karst conduits. The comprehensive review of karst hydrology by White (2002) suggested that sediment transport in karst settings remained one of the most unstudied aspects of karst in need of research. Since that time, a number of groups have investigated the ability of fluviokarst networks to transport sediment and have found that rainfall activated surface tributaries can carry high sediment loads and provide quickflow to the subterranean karst (Hart and Schurger, 2005; Massei et al., 2003); karst drainages entrain and transport sediment loads as function of fluid intensity, similarly to surface streams (Dogwiler and Wicks, 2004); and karst systems store and convey a distribution of sediment under varying ground saturation, moisture, and discharge conditions (Hart and Schurger, 2005; Herman et al., 2008). From recent sediment transport studies, an important feature has been the realization of a sub-classification of karst in phreatic systems. Phreatic conduits are situated below the water table and therefore have a downstream hydraulic control structure, i.e., subterranean dam, or adverse conduit gradient in the streamwise direction that produces saturated flow conditions. In terms of hydraulics, phreatic conduits have an upper limit for their energy gradient and thus upper limit for fluid conveyance due to the existence of the downstream controls. The fluid energy threshold of the phreatic conduits offers the potential to trap sediment either temporarily or permanently (Herman et al., 2008), which highlights the potential for sediment carbon mineralization within the fluviokarst system.

Advancement in our understanding of sediment carbon fate and hydrological processes in karst relies on the application of new or advanced instrumentation within karst systems as well as adopting existing methods from other fluvial settings and applying them to karst. Methods in karst have been greatly advanced in recent years, with a number of methods available for hydrologic analysis. Water conveyance methods generally consist of gaging stations for flow estimation installed at swallow holes and springs (Mahler and Lynch, 1999; Bonacci, 2001; Reed et al., 2010), piezometers for continuous measurement of the groundwater table (Long and Derickson, 1999), and natural as well as artificial tracers for understanding water origin and connectivity between surface and subsurface pathways (Katz et al., 1997; Perrin et al., 2003; Barbieri et al., 2005). Sediment measurements in karst aquifers are typically performed by scraping cave surfaces, pumping or coring at well sites, automated pump sampling at spring outlets (Mahler et al., 1999; Herman et al., 2008; Reed et al., 2010), and use of sediment fingerprinting techniques for distinguishing sediment sources and estimating residence time (Mahler et al., 1998; Pronk et al., 2006).

In the present paper, the authors apply the above mentioned data collection methods and also work to extend the karst scientific toolbox in order to understand sediment carbon fate. The authors apply carbon stable isotopes for understanding the source of sediment carbon supplied to the karst subsurface *via* swallets and for investigating the fate of carbon within the subsurface. The stable isotopic signature of carbon ($\delta^{13}C$) is inherently linked to the land use origin of sediment from different plant type and management scenarios (Fox and Papanicolaou, 2008) as well as to the organic matter structure of carbon due to its sensitivity to the level of microbial processing (Acton et al., 2013). Carbon stable isotopes have been previously used in fluvial environments for understanding the source and fate of sediment carbon as well as within sediment fingerprinting (Fox and Papanicolaou, 2007; Fox, 2009; Jacinthe et al., 2009; Mukundan et al., 2010; Ford and Fox, 2015; Fox and Martin, 2015). However, to the authors' knowledge, the method has not been applied in karst settings. In addition to the use of stable isotopes and traditional sampling methods, the authors install several monitoring wells which directly intersect the primary karst at its longitudinal midpoint in order to continuously monitor water and sediment. The authors find few studies in the literature that have continuously collected hydrologic data at karst inlets and outlets as well as from within the primary conduit draining the aquifer.

This study's objectives were to elucidate previously unstudied hydrological processes within phreatic karst and develop a conceptual model of sediment carbon fate within phreatic karst. The conceptual model is discussed in the context of active freshwater carbon cycles. Thereafter, the conceptual model is used as a guide to build a numerical model in our companion paper (Paper 2: Numerical Model) that immediately follows this article in this journal.

2. Methods

2.1. Conceptual model development

The authors focus their conceptual model development for sediment carbon in phreatic karst upon hydrologic and landscape features that provides a sub-classification of karst systems (see Fig. 1). The authors emphasize mature, phreatic karst systems with hydraulically connected surface water and subsurface water. Sinking streams and swallets located in the surface stream corridor are fluviokarst features that can transport stream sediment to subsurface conduits and caves. The authors focus on phreatic karst such that a subsurface hydraulic control has the potential to mediate fluid energy, cause trapping of sediments, and potentially allow for the mineralization of sediment carbon. The authors emphasize karst systems with active subsurface conduit flow that can convey sediment to a springhead. The existence of a springhead allows connectivity of sediment carbon back to the fluvial network, which highlights the broader goal of understanding karst landscapes within the fluvial carbon cycle. Many phreatic karst systems reported upon in the literature can be characterized by the features mentioned above and conceptualized in Fig. 1 (White, 2002; Drysdale et al., 2001; Massei et al., 2003; Herman et al., 2008), yet sediment carbon fate and transport is understudied in such phreatic systems.

With the mentioned hydrologic and geologic characteristics in mind, the authors chose a mature karst system to assist with the conceptual model development for sediment carbon in phreatic karst. The study site chosen is the coupled Cane Run Creek Watershed and Royal Springs Groundwater Basin located in the Bluegrass

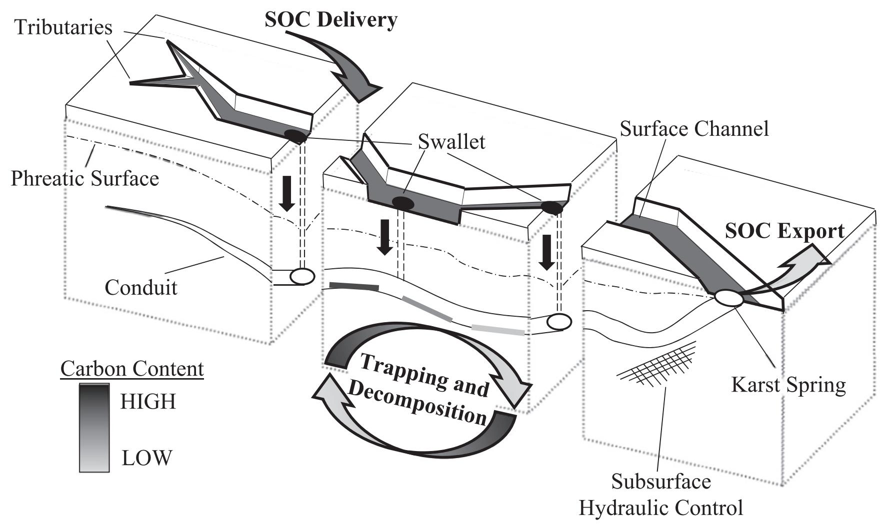


Fig. 1. Conceptual model of sediment organic carbon (SOC) transport in phreatic karst. Emboldened text indicates processes.

Region of central Kentucky, United States. Reasons for choosing the study site for development of our conceptual model were as follows: (1) The surface stream network of Cane Run Creek has high connectivity to the subsurface such that stream sediments can be conveyed to phreatic karst. Fifty-seven karst holes (e.g., swallets) have been mapped in and around the surface stream corridor, and many of these features connect surface water and sediment carbon to a primary, phreatic conduit (Taylor, 1992; Paylor and Currens, 2004). (2) The subsurface karst system consists of a series of anastomosis conduits that converge to the primary, phreatic conduit that transports water and sediment carbon. The phreatic conduit is approximately 20 m below the ground surface, 5.4 m^2 in cross-sectional area at its longitudinal midpoint, elliptical in cross-section (6 m wide by 0.9 m height), and gains 15 m of elevation from its low point to springhead due to a subsurface hydraulic barrier (Thrailkill et al., 1991). The existence of the active conduit allowed the authors to investigate how phreatic karst might convey, trap, and turnover sediment carbon. (3) The phreatic conduit recharges water and sediment at a springhead allowing connectivity of sediment carbon back to the fluvial network. The Royal Spring springhead has the largest base flow discharge of any spring in the region and conveys perennial flow from the phreatic conduit (Currens et al., 2015). (4) The Cane Run-Royal Springs system was also chosen due to the large amount of previous morphologic and hydrologic study of the basin (Spangler, 1982; Thrailkill et al., 1991; Taylor, 1992; Paylor and Currens, 2004; Currens et al., 2015). (5) Finally, the karst system was chosen due to its close proximity of 15 km to the University of Kentucky and Kentucky Geological Survey headquarters allowing researchers to easily access the site throughout the course of this study.

2.2. Methodological approach

The authors' methodological approach for developing a conceptual model for sediment carbon in phreatic karst first relied on mapping the subsurface phreatic karst morphology as well as karst inlets and outlets for the specific system studied. Next, the authors sampled water and sediment carbon within the subsurface phreatic conduit, and the authors sampled water and sediment carbon entering and exiting the subsurface phreatic karst. Thereafter, the authors used analyses of the data streams and data-driven mass balances (see Fig. 2) to elucidate hydrological processes within the phreatic karst. The authors then infer sediment carbon fate within the study system that might be characteristic of a conceptual model of sediment carbon in phreatic karst more generally.

The authors mapped the subsurface phreatic karst morphology and its connectivity with the surface streams using 37 electrical resistivity profiles analyzed with the dipole-dipole electrode configuration method to estimate the extent of the primary conduit (Zhu et al., 2011). 44 wells were drilled to 20–30 m in depth to intersect and map the primary phreatic conduit, and potentiometric surface mapping was performed to estimate flow direction within the fracture aquifer to the conduit. Field investigation of swallet and springhead morphology was performed to measure inlets and outlets. Underwater camera observation and Doppler sonar techniques were used to estimate the phreatic conduit geometry.

The authors sampled water and sediment carbon within the subsurface phreatic conduit and at subsurface inlets and outlets at the stations shown in Fig. 3 for the coupled Cane Run Creek Watershed and Royal Springs Groundwater Basin. In the figure, it is shown that surface stream network conveys water and sediment carbon from urban and agricultural land surfaces to Cane Run Creek, which flows in the northwestern direction. The surface water and sediments are pirated via the 57 sinking streams and swallets to the phreatic conduit. The phreatic conduit is north to northwest flowing to Royal Springs, and its groundwater basin is shaded in Fig. 3. The subsurface phreatic conduit drains the landscape year round while the main stem of Cane Run Creek is only active about 10% of the year. During periods of high intensity or long duration precipitation, surface water and sediments overtop the swallets and continue downstream as surface flow.

The authors used their understanding of the karst system to choose stations for sample collection. Water and sediment carbon entering from the surface streamflow to the subsurface phreatic conduit were monitored at streamflow stations including the Agricultural Surface Flow Station and Urban Surface Flow Station. The streamflow stations were representative of urban and agricultural streamflow, in general, for the basin because: the urban or agriculture land-use dominated the drainage area; and the streams stations were located upstream of the swallets or sinking streams. The Surface Outflow Station was monitored to sample water and sediment carbon that overtops the swallets and sinking streams during high flow events and thus exits the watershed via surface flow. The phreatic karst conduit was directly monitored within the conduit near its longitudinal midpoint at the Groundwater Station and at is springhead at Royal Springs Station. Three wells

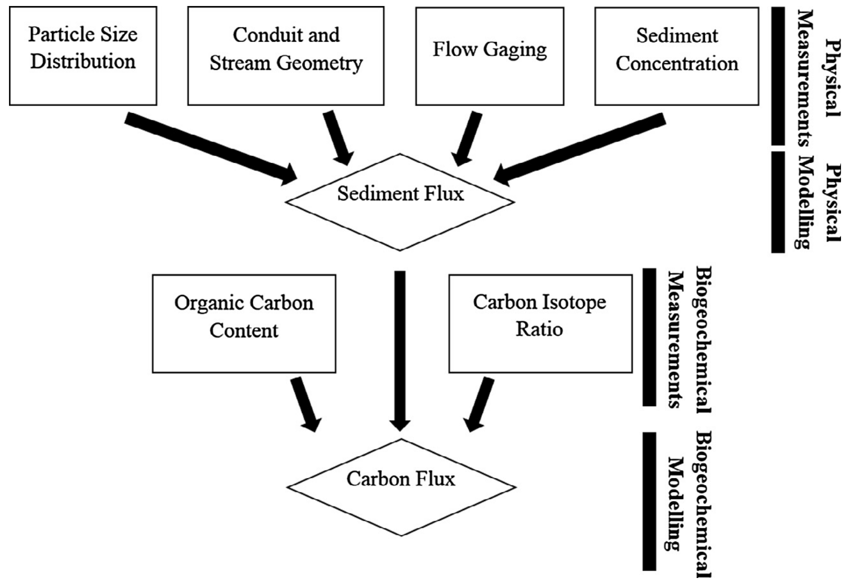
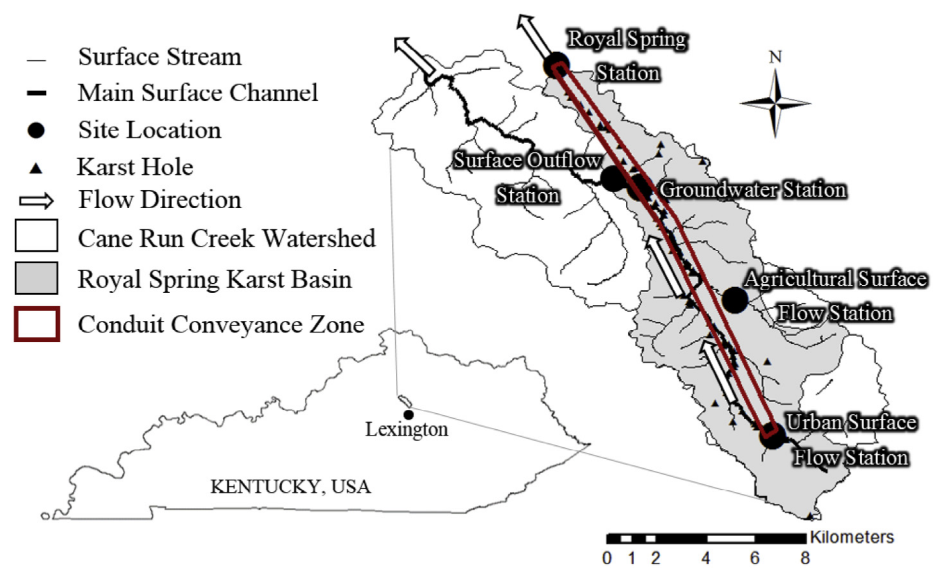


Fig. 2. Diagram of the methodological approach for the fluviokarst sediment organic carbon study.

a)



b)

c)

Attribute	Description
Geology	Lexington Limestone of the Middle Ordovician period[1]
Rock structure	Minor faults and joints directing from the SE towards the NWt[2,3]
Topography	Rolling hills with fluviokarst features (e.g., swallets)[4,5]
Land use	38% urban and 62% agricultural
Soil	Silt loam and silty clay loam formed from weathered phosphatic limestone[6]
Climate	Averages 115 cm of rainfall per year and 13.1°C MAT[7]

[1]Cressman and Peterson, 1986; [2]Drahovzal *et al.*, 1992; [3]Drahovzal and Noger, 1995; [4]Paylor and Currens, 2004; [5]Husic, 2015; [6]USDA, 1993; [7]Bluegrass Regional Airport

Fig. 3. (a) Cane Run watershed and Royal Spring basin, (b) karst swallet pirating surface flow during low flow, and (c) watershed attributes. (See above-mentioned references for further information.)

directly intersected the phreatic conduit at the Groundwater Station, which allowed for sample collection.

2.3. Continuous water and sediment monitoring

The authors designed continuous water and sediment monitoring with emphasis upon elucidating fluid energy and sediment transport within a conceptual model for sediment carbon in phreatic karst. To do so, the authors carried out continuous flow monitoring at all five sampling stations shown in Fig. 3 at a 10 min sampling rate for two years (1 October 2011 to 30 September 2013). Extensive details of the quality assurance protocol for sample collection is provided in Husic (2015), and the primary method applied is included herein. The surface stream stations were instrumented with *in situ* pressure transducers, and velocity measurements were collected at different stages to develop stage-discharge relationships for each station. Instrumentation installed at the Groundwater Station included a permanent Marsh-McBirney 201-D continuous velocity recording device as well as several Telog 2109 Water Level Recorders. The velocimeter was placed at 80% of the height of the conduit to collect the depth average velocity as estimated by the one-seventh power law (De Chant, 2005). At Royal Spring Station, the United States Geological Survey (USGS) operates a v-notch weir and associated staff gage (USGS 03288110). Water discharge estimates were used to calculate a data-driven water budget (Table 1) in which the agricultural and urban stream stations were scaled to represent inputs to the basin.

Temperature measurements and dye traces assisted with understanding hydrologic connectivity within the phreatic karst system. Temperature data were recorded with YSI 6920v2 water quality sondes at the sampling stations to monitor the flushing of pre-event conduit water by quickflow from the surface. Quantitative and qualitative dye traces were used to estimate travel time and swallet connectivity to the phreatic conduit. Rhodamine WT and fluorescein were injected into a karst window at a travel distance of approximately 1.5 km upstream from the Groundwater Station following established methods (Smart and Laidlaw, 1977; Wilson et al., 1986). Downstream tracer concentration was measured by collecting water samples every 10 min using a Teledyne ISCO 6712 pump sampler. Fluorescein tracer analysis was performed with a Cary Eclipse Varian fluorescence spectrophotometer. The arrival time of the center of mass of the fluorescein was used to estimate the velocity of the flow. Additionally, a conservation of mass approach was applied to the Rhodamine WT dye trace in order to estimate conduit discharge (Gouzie et al., 2015).

Sediment measurements were coupled with the water measurements to estimate particle size characteristics and sediment discharge at the five stations. Suspended sediment was collected at the sampling stations and analyzed using a LISST-Portable|XR to estimate particle size distribution. The method for estimating sediment transport rates at the stations applied the sediment concentration relationship for tributaries in the region coupled with Einstein's Approach, which integrates the velocity and sediment concentration profiles (e.g., Chang, 1988; Russo and Fox, 2012). Velocity profiles relied on the modified logarithmic law and one-seventh power law for the streams and conduits, respectively (Chang, 1988; De Chant, 2005). The friction velocity in the streams and conduits was estimated using the momentum equation and Darcy-Weisbach equation, respectively (Chang, 1988, pp. 41; Allen et al., 2007; Husic, 2015). Continuous data were input to the sediment discharge formula at 10 min intervals, and data input included sediment concentration, water depth in the stream, and velocity within the conduit (Husic, 2015). Sediment concentration measurements were measured using water samples collected with Teledyne ISCO 6712 pump samplers, and then continuous estimates were provided by coupling concentration measurements with continuous YSI 6920v2 turbidity probe measurements, which is commonly performed for sediment budget studies (e.g., Walling et al., 2006).

Table 1
Water budget for coupled surface-subsurface watershed presented in inches of rainfall per year normalized by catchment area.

Input/Output	($cm\ km^{-2}\ y^{-1}$)
Agriculture Tributaries	12.2
Urban Tributaries	39.9
Surface Outflow	11.9
Groundwater Station	37.8
Change in Volume	2.4

2.4. Sediment carbon monitoring

The authors designed sediment carbon monitoring with the conceptual model development for sediment carbon in phreatic karst in mind, and specifically focused on carbon sources to the phreatic karst and carbon fate within the phreatic conduit. As a first step, the authors applied sediment carbon concentration and stable carbon isotope ($\delta^{13}C$) measurements to fingerprint the sources of sediment carbon entering the phreatic conduit. As a second step, the authors applied carbon and $\delta^{13}C$ measurements to estimate the fate of sediment carbon within the phreatic conduit by analyzing data input from the surface streams and output from the phreatic conduit at the springhead.

The authors considered carbon sources entering the phreatic conduit by recognizing that urban and agricultural surface streams transport sediment carbon derived from terrestrial and aquatic origin within the fluvial load (<53 μm in diameter) (Arango et al., 2007; Cole et al., 2007; Trimmer et al., 2012; Ford and Fox, 2014). Thus, sediment carbon is a mixture of: (i) a terrestrial carbon pool that includes fine-sized litter and newly derived soil carbon from litter or root turnover; (ii) a terrestrial carbon pool of recalcitrant soil carbon that has undergone numerous stages of decomposition; and (iii) an aquatic carbon pool of disaggregated and humified algae produced in the bed of the stream network (Ford et al., 2015). The three carbon sources are worthy of note because they will vary in their recalcitrance (Cambardella and Elliott, 1992; Marwick et al., 2015), and will provide ecosystem energy production, and hence carbon turnover, *via* oxidation by heterotrophic bacteria (Thorp and Delong, 2002). Fine-sized litter is high in carbohydrates with high C:N ratios, whereas older soil carbon has a high contribution of microbial processed and synthesized compounds with smaller C:N ratios (Marín-Spiotta et al., 2014). In turn, more highly bioavailable carbon within labile litter will provide high energy production per unit mass of carbon relative to the older, more recalcitrant pool (Thorp and Delong, 2002). Studies of *in situ* organic matter decomposition in streams suggest that sediment carbon recently derived from leaf litter and detritus has decomposition rates on the order of $1 \times 10^{-3}\ d^{-1}$ while older soil carbon has decomposition rates on the order of $1 \times 10^{-5}\ d^{-1}$ (Webster et al., 1999; Six and Jastrow, 2002; Yoshimura et al., 2008). Algal-derived sediment carbon is recognized as a carbon-rich pool composed of highly labile neutral sugars (Vieira and Myklestad, 1986; Waite et al., 1995; Lane et al., 2013) and, in turn, will have decomposition rates on the order of $1 \times 10^{-3}\ d^{-1}$ or higher (Ford and Fox, 2015).

The authors applied carbon fingerprinting for estimating the contribution of litter, soil, and algal carbon to the phreatic karst using tracer un-mixing (Davis and Fox, 2009) as

$$y^T = \sum_k (x_k^T \times P_k), \quad (1)$$

and

$$\sum_k P_k = 1, \quad (2)$$

where, y is the tracer of sediment carbon collected from the mixture location in the stream, x is the tracer of a carbon source, T designates an index for the tracer being used, k designates an index for the carbon source, and P is the mass fraction of carbon originating from a particular source. In the present analysis, the stable carbon isotope (δ^{13}C) of sediment carbon was chosen as the biomarker tracer to un-mix the carbon pools. δ^{13}C is inherently linked to the organic matter structure of the carbon pool (Sharp, 2007) and has been found to discriminate terrestrial carbon and aquatic pools so long as the nature of the carbon pool and end-members are properly characterized and δ^{13}C is treated as conservative (Ford and Fox, 2015; Fox and Martin, 2015). In the present study, urban tributaries are storm event-activated and do not sustain flow necessary for primary in-stream production hence only two sources (i.e., soil and litter) were considered for tributaries draining urban lands. Sediment carbon fingerprinting from agriculture tributaries contained all three sources.

As mentioned, the second step of the sediment carbon monitoring focused on estimating the fate of carbon within the phreatic conduit. The microbial decomposition of carbon was estimated during temporary storage as

$$SOC_{out} = SOC_{in} - DECt_R, \quad (3)$$

where Eq. (3) is a first-order carbon turnover model commonly applied for carbon cycling in freshwater (Shih et al., 2010; Ford and Fox, 2014). SOC_{in} is the sediment carbon composition of sediment entering the subsurface karst from the surface streams (g C), SOC_{out} is the sediment carbon exiting the subsurface karst at the springhead (g C), DEC is the net microbial decomposition rate that can be estimated when the distribution of carbon sources to the conduit is known or estimated (g C d^{-1}), and t_R is the net residence time of the sediment carbon in the conduit (d). It was recognized that influx of sediment carbon into the karst system is likely episodic and driven by the occurrence of hydrologic events, and, for this reason, the net residence time estimated in Eq. (3) assumes equilibrium over several years and relies on repetition of samples to estimate mean sediment carbon concentrations entering and exiting the phreatic karst.

In addition to the net change in sediment carbon concentration within the phreatic conduit, the authors considered the change in the stable carbon isotopic signature of sediment. The authors assumed long term equilibrium and applied a Rayleigh-like fractionation model (Ford and Fox, 2016) as

$$\delta^{13}C_{out} = \delta^{13}C_{in} - \varepsilon \ln f, \quad (4)$$

where the carbon isotope signature is changed *via* the product of enrichment via fractionation during decomposition, ε (‰), and the natural logarithm of the net organic carbon lost during decomposition, *f*. The enrichment factor associated with decomposition of fine sediment carbon is on the order of 0–2‰ (Jacinthe et al., 2009). Eq. (4) provides an independent method to assess aerobic microbial decomposition of sediment carbon given carbon concentration and δ^{13}C of sediment carbon entering and existing the karst subsurface.

To carry out the sediment carbon source and fate analyses in Eqs. (1), (3) and (4), transported sediment carbon was collected from the surface flow stations and the springhead station using *in situ* sediment trap samplers over the course of 22 months. The sampling method relied on the use of time-integrated sediment samplers, which have been found to provide a representative, integrated total carbon signature for a stream (Phillips et al., 2000; Fox and Papanicolaou, 2007; Ford and Fox, 2014; Fox et al., 2014). Sediment traps were installed at the sampling stations and samples were collected from the traps on a weekly basis. Samples collected from traps which were clogged and samples with an inadequate sediment mass were not included in the analysis in order to avoid biasing. Samples were processed back in the laboratory following the methods outlined in Ford and Fox (2014) and Husic (2015). In brief, the samples were dewatered and weighed, wet-sieved through a 53 µm sieve, dewatered and weighed again, ground to a fine powder, and acidified repeatedly using 6% sulfurous acid (Verardo et al., 1990). Sediment carbon samples were analyzed for elemental and isotope composition by combusting samples at 980 °C on a Costech 4010 Elemental Analyzer, passing the gas stream through a Gas Chromatograph (GC) column (3 m HS-Q) to a Thermo Finnigan Delta-Plus XP Isotope Ratio Mass Spectrometer (IRMS). The carbon elemental signature, *C*, was reported as a percentage of the mass of carbon relative the mass of sediment. Isotopic results were reported in delta notation as

$$\delta^{13}C = \left(\frac{R_{Sample}}{R_{Standard}} - 1\right) * 1000 \quad (5)$$

where R_{Sample} is the $^{13}C/^{12}C$ ratio of the samples and $R_{Standard}$ is the $^{13}C/^{12}C$ ratio of the universal standard, Vienna Pee Dee Belemnite (VPDB). The elemental reference was acetanilide (%C = 71.09%), and isotopic references were DORM ($\delta^{13}C = -19.59$), and CCHIX ($\delta^{13}C = -16.4$‰). Average standard deviations for elemental and isotopic standards were 0.34% and 0.20‰, respectively. Average standard deviations of replicates were 0.10% and 0.08‰ for carbon concentration and $\delta^{13}C$, respectively.

Carbon isotope signatures applied in Eq. (1) were previously collected from litter, soil, and algae stocks in nearby Kentucky watersheds with similar lithologic, soil, C3 plant type, and benthic algae characteristics (Fox et al., 2010; Acton et al., 2013; Ford et al., 2015). Within the carbon fingerprinting analysis, Eqs. (1) and (2) were under-parameterized for the condition of a single tracer, and therefore additional field information was integrated into the analysis. The source fraction of algae contributing to sediment carbon in the surface streams was estimated using the results from nearby streams in the Inner Bluegrass (Ford et al., 2014). The average value of percent algae in these streams was found to be 17.8% (Ford et al., 2014), and the authors in this study varied this range widely from 0 to 40% algae to account for uncertainty within the results.

3. Results and discussion

As will be shown, the authors used analyses and interpretation of the data streams to elucidate understudied hydrological processes within phreatic karst including the sediment transport carrying capacity of the flow during and after storm events and the functioning of the surficial fine grained laminae. Thereafter, the authors discuss a conceptual model that may be characteristic of sediment carbon in phreatic karst more generally whereby phreatic karst temporarily stores sediment, turns over carbon at higher rates than would be considered otherwise, respires carbon dioxide to the water column, and recharges degraded organic carbon back to the surface stream.

3.1. Water conveyance in phreatic karst

Water and its conveyance provides the medium and energy by which sediment carbon is transported, stored, and turned over in phreatic karst. Numerous studies have presented results of water conveyance in phreatic karst. Therefore, the authors recognized the need to measure hydrologic connectivity and response time of their study system to who that it behaves similarly with phreatic karst systems for which the conceptual model of sediment carbon is sought after.

Surface streams in the study area were event activated exhibiting high stormflow and low baseflow periods. The urban stream was generally much more active with regards to storm flow than the agricultural stream (Fig. 4), which was attributed to the higher percentage of impervious areas contributing runoff. The Surface Outflow Station displayed a similar behavior as the urban and agricultural streams in that it was active primarily during hydrologic events and had relatively short-lived hydrographs. Peak stormflow in the surface streams was orders of magnitude greater than baseflow (Fig. 4).

Water conveyance results were quite different for the phreatic conduit in comparison to the surface streams (see Groundwater Station in Fig. 4). The phreatic conduit exhibited sustained year-round flow, but flow was buffered due to limited conveyance of the subsurface pathway. Well stage data from the conduit and surrounding karst aquifer showed that even during very low flow conditions the conduit remained phreatic; the groundwater table fluctuated 6–16 m above the mid-point of the conduit. The mean conduit velocity was 0.12 m s^{-1} and the standard error was small (±0.11), especially relative to surface streams. Fig. 4 shows that peak flows in the conduit were limited in their extremes relative to the surface streams. Flow rate in the surface streams was as high as 25 m^3 s^{-1} while flow in the conduit was an order of magnitude lower and never exceeded 3 m^3 s^{-1}. The limited water conveyance was attributed to the dimensions of the karst conduit (i.e., 0.9 m $\times$ 6 m), the downstream pressure gradient induced by the hydraulic control, and intermittent swallet overflow. The sustained perennial flow of the conduit resulted in 76% of the water that exited the coupled surface-subsurface system occurred via the phreatic conduit.

Water conveyance time-series measurements suggested confidence that the surface streams and phreatic conduit have high hydrologic connectivity that would allow for active sediment carbon delivery to the subsurface karst. Temperature and discharge time series from a storm event in March 2013 show the temperature response of the conduit at the Groundwater Station to quickflow from surface tributaries during the rising limb of the hydrograph (Fig. 5). Water temperature decreases at the Groundwater Station before flow is recorded at the Surface Outflow Station indicating that initial surface flows are pirated before continuing downstream and reaching the surface outlet. The temperature decrease at each location occurs within the first 18 h of the event indicating the close coupling of surface streams and phreatic karst. Additional justification for the high connectivity between the surface and subsurface was provided by the fast travel times within the conduit estimated from dye traces (Table 2). The average travel time from the dye traces was scaled to the entire conduit, and it is estimated that fluid travels a distance of 16 km over approximately 22 (±6.8) hours. Results highlight the relatively high velocity of fluid (20 cm s^{-1}) and hydrologic connectivity of the surface-subsurface system during hydrologic events.

The results of water conveyance in the study system are consistent with the features of phreatic karst for which a conceptual model of sediment carbon is sought after (see Fig. 1) and agree with phreatic karst hydrology reported in other studies. For example, mature karst morphology is well-recognized to have conduit networks developed along geologic bedding planes with water at velocities orders of magnitude greater than porous media or fracture matrix flows (Atkinson, 1977; White, 2002; Waltham and Fookes, 2003). A number of studies have suggested that recharge occurs to phreatic conduits during stormflow when rainfall activated surface water tributaries carry quickflow *via* swallets to the subterranean karst (Vesper and White, 2004; Massei et al., 2006). The finite water conveyance of karst conduits and caves due to internal energy controls and springhead overflow has long been identified in karst literature (White, 1988; Bonacci, 2001).

3.2. Sediment transport in phreatic karst

The hydrologic connectivity of surface streams to swallets to phreatic conduits to springheads back to surface streams coupled with phreatic water conveyance that is buffered during storm events yet sustained perennially suggests a particularly 'jerky

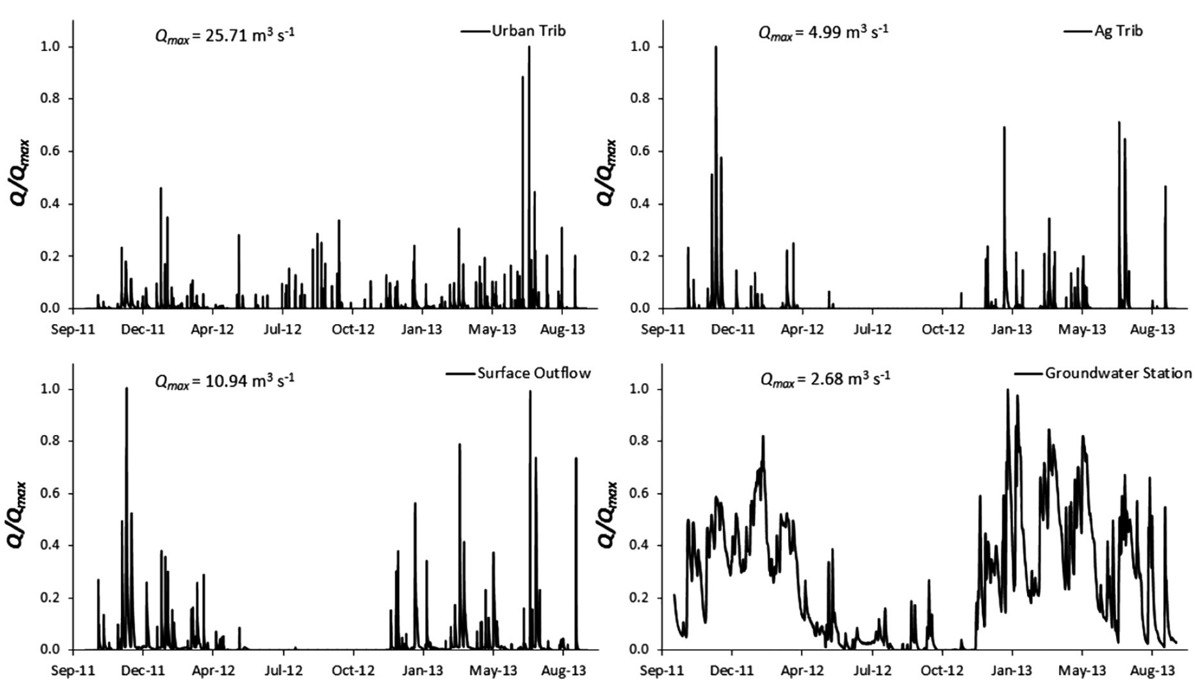


Fig. 4. Inflows and outflows to the watershed normalized by maximum flow rate at each respective location.

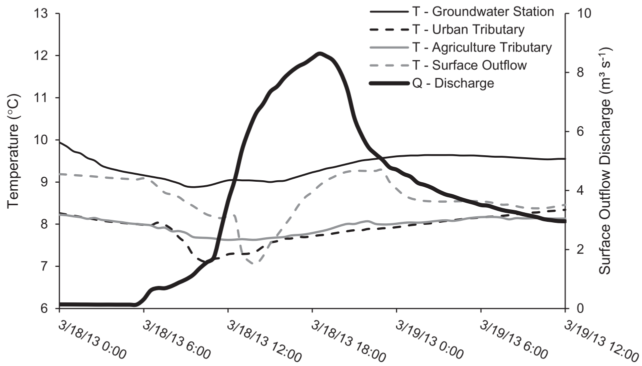


Fig. 5. Temperature fluctuations during a storm at four sampling sites. Discharge shown on figure is from the Surface Outflow site.

Table 2
Dye trace experiment results from an eclipse karst window to the Groundwater Station (1.5 km).

Date and Time	Dye Spike Velocity (m s^{-1})	Travel Time (hr)
6 Dec 2011 14:50	0.32	1.3
1 Mar 2012 14:00	0.15	2.7
10 Dec 2012 15:40	0.20	2.0
7 Feb 2013 18:00	0.14	2.9
12 Mar 2013 13:20	0.18	2.3
19 Mar 2013 11:35	0.28	1.5
12 Apr 2013 15:30	0.14	2.9
8 Apr 2014 13:00	0.18	2.3
Average	0.20	2.2
Standard Deviation	0.07	0.6

conveyor' for sediment within phreatic karst (as coined by Ferguson (1981), for fluvial systems). The sediment residence time in fluvial systems that includes phreatic karst pathways is expected to be increased relative to surface-dominated systems. The authors use their data results and literature comparison in this section to elucidate hydrologic processes in phreatic karst including the role of the sediment transport carrying capacity to induce deposition, temporary storage, and resuspension of sediment carbon as well as the presence of the surficial fine grained laminae. In turn, the results lead to a conceptual model suggesting that phreatic karst turn over carbon at higher rates than would be considered otherwise in fluvial systems.

Sediment discharge results (Fig. 6) show that sediment transport near the longitudinal center of the phreatic conduit has low sediment transport rates during hydrologic events relative to the surface streams that input sediment to the subsurface. For example, the peak sediment concentration and discharge within the phreatic conduit was 192 mg L^{-1} and 0.27 kg s^{-1}, respectively, which were substantially smaller than the urban surface stream sediment concentration and discharge of 1584 mg L^{-1} and 29.73 kg s^{-1}, respectively. The sediment transport rate differences between the phreatic conduit and surface streams are not explainable based on particle size differences. Particle size results suggest that very little sorting occurs during the transport process as the particle size distribution of conduit suspended sediments nearly match the particle size distribution of suspended sediments in the surface streams (Fig. 7). Rather, the results are explained based on deposition of sediment within the phreatic conduit. The surface streams input water with high sediment concentration directly to the karst swallets, however, results from the Groundwater Station suggest that the majority of the sediment has fallen out of suspension by the time the water reaches the longitudinal center of the conduit. As mentioned, water flow results suggest the water travel time is 22 (±6.8) hours, which provides ample time for settling considering the settling velocity and conduit height that provides a deposition time of approximately 0.14 h ($t = 0.5\,Hw_s^{-1}$).

Downhole imagery of the phreatic conduit provided justification of pronounced sediment deposition within the phreatic conduit. Fine sediment was present on the conduit bed along with larger limestone rocks that also were covered with a layer of fine sediment. In frames of the video, suspended sediment transport was also visually observed within the conduit moving at relatively high velocities. Blanketing of the cave's floor with a fine sediment layer is consistent with fluvial sediment entering and exiting the conduit and suggests deposition of transported fine sediment. This fine sediment layer in fluvial systems has been termed the surficial fine grained laminae (Droppo and Stone, 1994) and is recognized to be active both physically in terms of deposition and resuspension and biologically in terms of microbial growth and carbon turnover (Russo and Fox, 2012; Ford and Fox, 2014).

While sediment peaks during the hydrologic events are much smaller, sediment data results show that turbidity spikes in the phreatic conduit last through the peak of a hydrologic event and are maintained for much longer durations in comparison to the surface streams. From analysis of four characteristic hydrologic events, Fig. 8 shows that elevated sediment discharge in the conduit lasts approximately 2.5 times the duration of peak urban stream transport. The relatively high sediment concentration within the conduit continues to occur after input of sediment from tributaries has ceased. The results highlight that sediment transport occurs after the external sediment source has been cutoff. In this manner, conduit internal sediment deposited during the hydrologic events provides a sustained source in the absence of hydrologic events. The result occurs because water flowrates in the conduit are sustained for days to weeks after the storm event, and in turn the water conveyance provides fluid energy to erode conduit bed material and transport sediment to the springhead.

The deposition of sediment during storm events within the phreatic conduit, presence of the active surficial fine grained laminae, and later resuspension of sediment long after the storm pulse has passed through the surface streams can be well explained by considering the energy of the fluid to carry sediment. The sediment transport carrying capacity (T_C) of the flow was normalized by its maximum (see Fig. 9) as

$$\frac{T_C}{T_{Cmax}} = \frac{k_{tc}V^3}{k_{tc}V^3_{max}} \tag{6}$$

where, k_{tc} is a transport coefficient and V is the flow velocity (m s^{-1}). Analysis of the ratio of the surface stream transport carrying capacity to that of the conduit shows that during hydrologic events the sediment carrying capacity of the surface streams is many orders of magnitude greater than that of the conduit (i.e., $T_C^{urbanstream}/T_C^{conduit} = 10^3$). The result highlights the reason as to why pronounced deposition occurs in the conduit during hydrologic events and reinforces the limiting of sediment transport by the phreatic conduit. The transport capacity of the conduit is shown to be highly sustained relative to the surface streams (Fig. 9), which highlights the ability of water conveyed within the conduit to erode and transport sediment long after the surface hydrologic activity has ended. Surface events have short-lived transport capacity peaks with a subsequent return to low- or no-flow. The transport capacity within the karst conduits recedes much more slowly and is maintained for weeks after an event, i.e., water is continually supplied to the conduit by fractures, macropores, and the epikarst allowing for continued subsurface sediment transport. The result diverges the phreatic karst from surface streams and non-phreatic karst where the energy of the fluid is a function of flow depth. The sustained transport capacity promotes resuspension of fine sediments

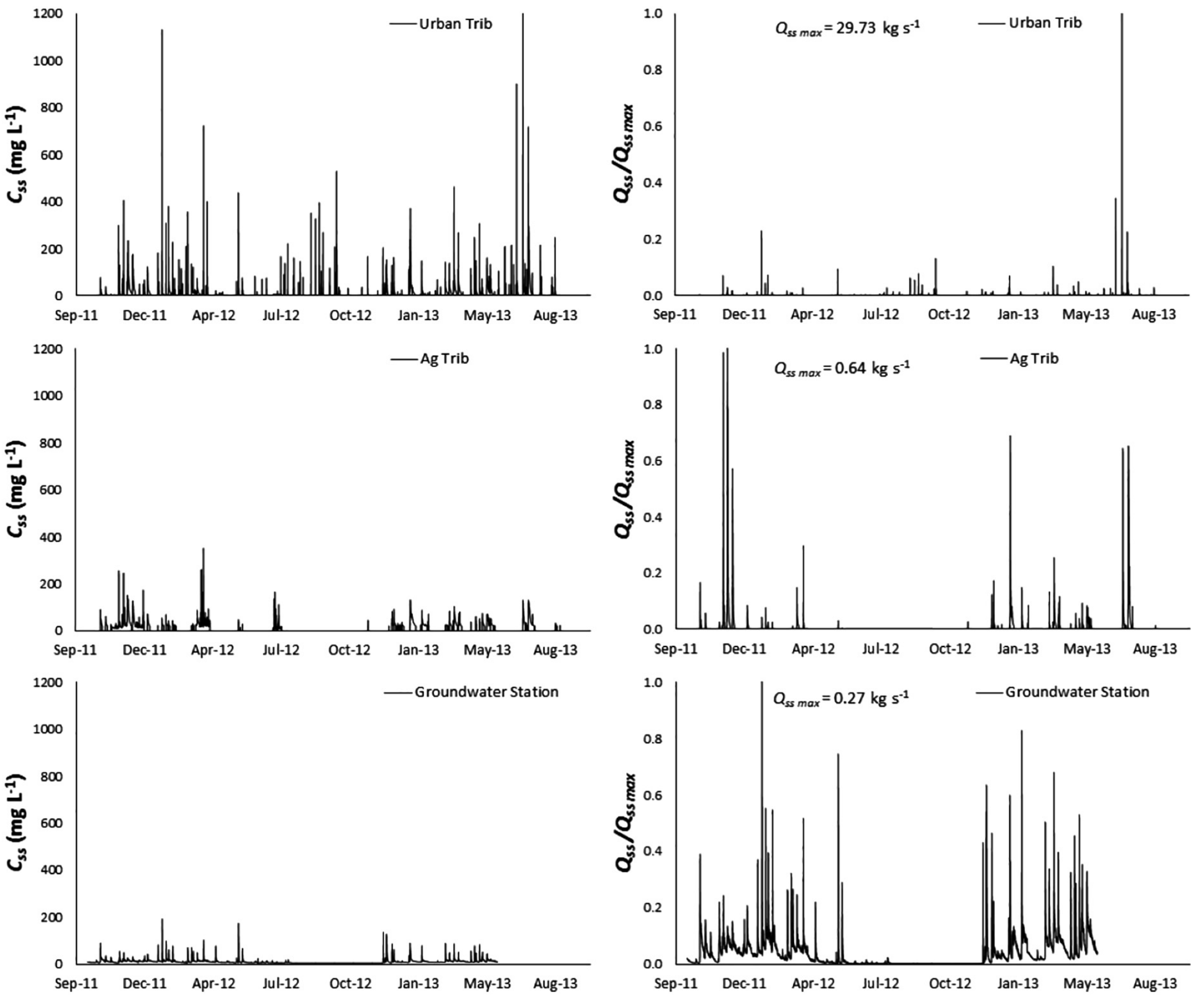


Fig. 6. Suspended sediment concentration at tributaries and Groundwater Station on the left. Suspended sediment discharge normalized by maximum sediment discharge at each location on the right. *Gap in Groundwater Station data starting on May 20, 2013 due to instrument failure.

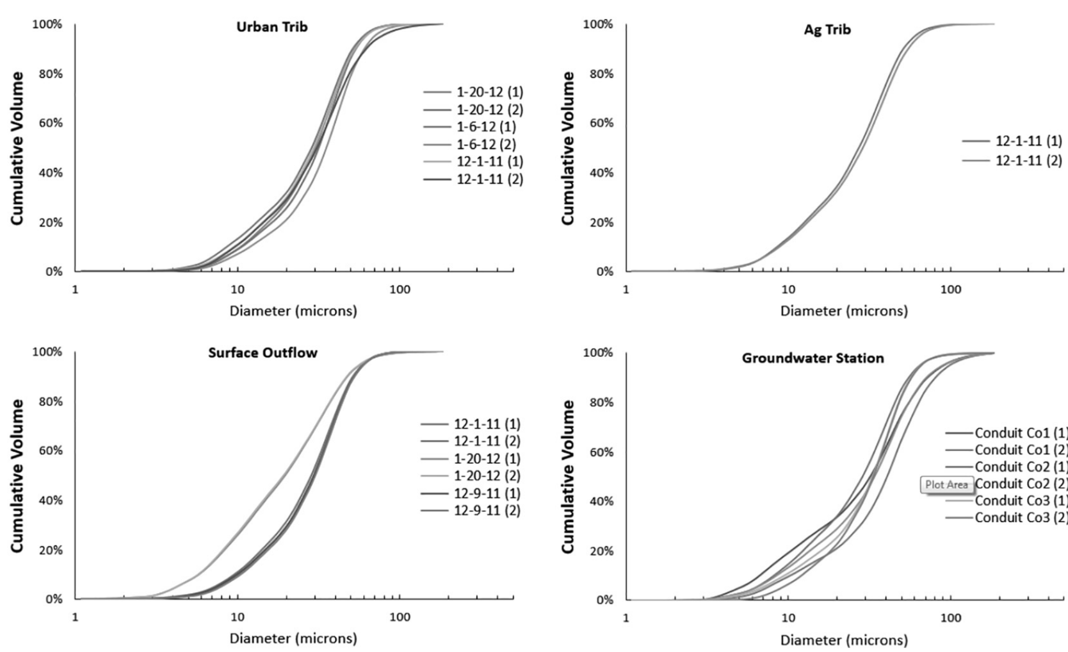


Fig. 7. Particle size distribution at inflows and outflows to the watershed.

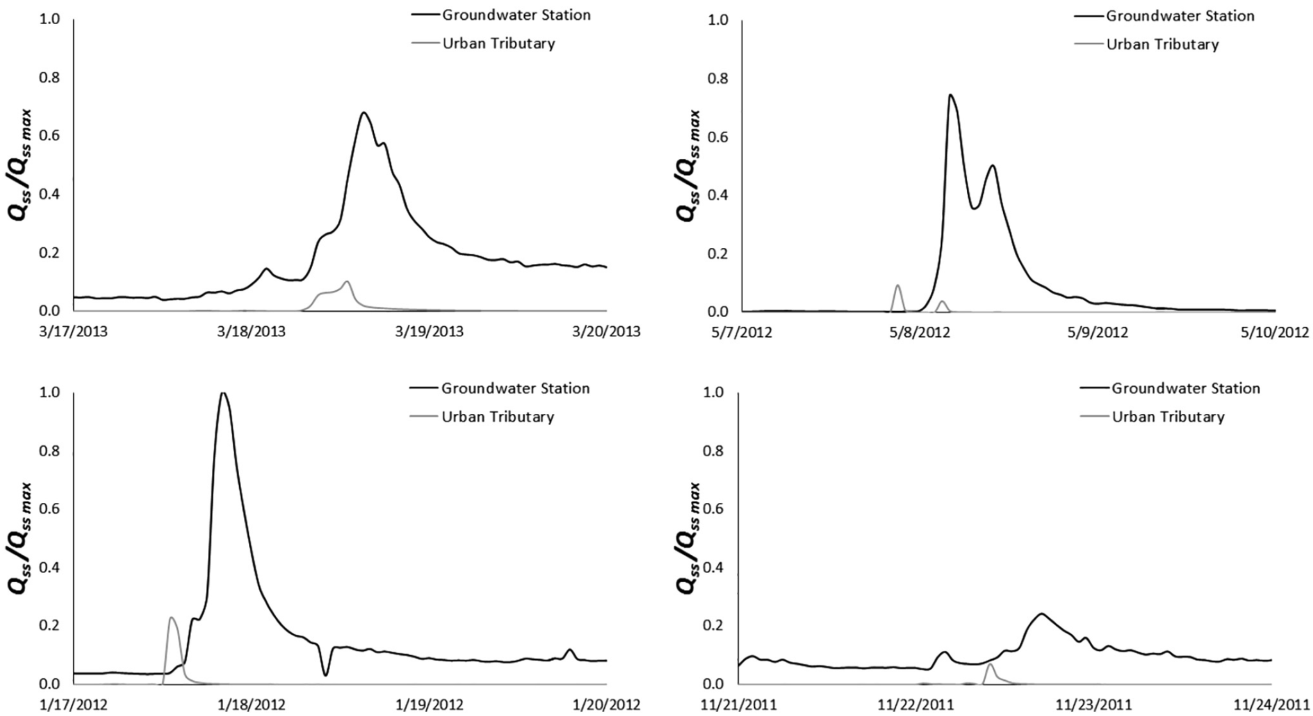


Fig. 8. Suspended sediment flux comparison for four characteristic hydrologic events. Note that the y-axis is normalized discharge; the urban tributary discharges a larger magnitude overall.

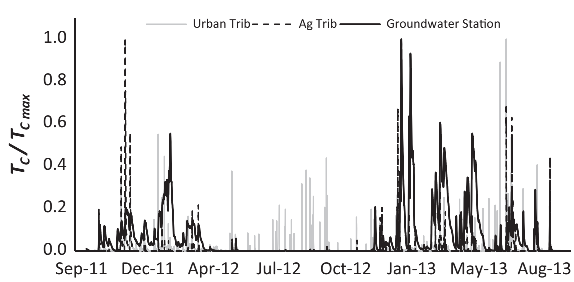


Fig. 9. Transport carrying capacity of urban and agricultural tributaries and the conduit normalized by maximum transport carrying capacity at each respective location.

long after surface events and sediment transport in the phreatic karst is higher than surface streams for most of the year (Fig. 9).

Hydrologic processes of deposition, temporary storage, and resuspension are discussed or alluded to in other phreatic karst studies and therefore provides further support towards our conceptual model for sediment carbon. For example, the fluid energy threshold of phreatic conduits has been suggested as a means to trap sediment (Herman et al., 2008). Specifically, the buffering and maintaining of the transport capacity resulting in deposition within the subsurface conduit during an event has been an observed phenomena by large sediment pulse deposition in a cave after hydrologic events (Gillieson, 1986), and hourly sampling of large and small hydrologic events showed prolonged high sediment loads at a springhead following a storm (Mahler and Lynch, 1999). Finally, the idea of an active surface fine grained laminae has been highlighted by the observance of epilithic biofilms in karst streams that have shown active microbial and invertebrate communities that turnover surface-derived organic matter (Simon et al., 2003) and carbon balances have shown the oxidation of sediment organic carbon during transport (Alberic and Lepiller, 1998).

3.3. Sediment carbon fate in phreatic karst

The hydrologic processes identified for phreatic karst point towards a conceptual model for sediment carbon that includes temporary storage, turnover of carbon at higher rates than would be considered otherwise, respiration of carbon dioxide to the water column, and recharge of degraded organic carbon back to the surface fluvial system. Such a conceptual model might be expected for karst that includes hydrologic connectivity and active sediment delivery from surface streams to the subsurface, the presence of phreatic conduits, and active recharge of back to the fluvial network.

Sediment carbon results from the present study support the conceptual model for sediment carbon. Surface stream sediment carbon input to the karst averaged 4.8 (±1.2) gC 100 g^{-1} sediment while sediment carbon collected from the conduit discharge averaged 3.4 (±0.5) gC 100 g^{-1} sediment. Some point data overlap existed for sediment carbon inflowing to and outflowing from the karst conduit (Fig. 10), which is at least partially attributed to suspended sediment that is flushed through the conduit during a hydrologic event. Carbon inputs and outputs were significantly different (p-value $< 1 \times 10^{-6}$) based on two-tailed statistical t-tests.

On average, results of carbon measurements show a 30% loss of sediment carbon when comparing inputs to the karst conduit with outflowing sediment at the springhead. The carbon density differences suggest that the temporary sediment carbon storage within the bed of the conduit promotes carbon turnover by heterotrophic bacteria. The explanation is reasonable given that the sediment carbon inflowing to the karst subsurface includes labile carbon pools and the karst water in the conduit is oxygenated and maintains a relatively constant water temperature. Sediment carbon within the surface streams that enters to the karst conduit *via* the swallets was found to be a mixture of fine-sized litter carbon, algae-originated carbon, and soil carbon (Table 3). Litter and algal carbon are recognized to be fairly labile carbon pools, and the labile

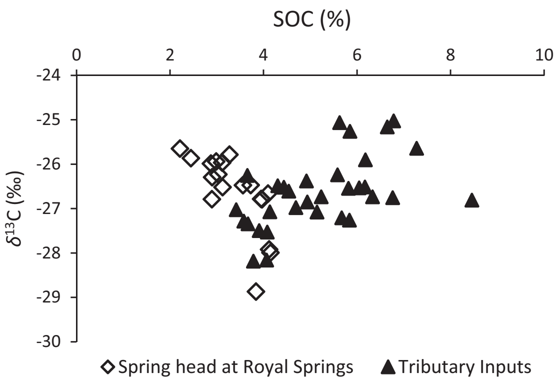


Fig. 10. Organic carbon content and carbon isotope values for inflowing tributary (n = 32) and outflowing spring (n = 18) sediment.

pools comprised approximately 50% of the total sediment carbon entering the conduit from urban waters and 50–75% from agricultural waters. Studies of *in situ* organic matter decomposition in streams suggest that particulate organic matter recently derived from leaf litter and algae have decomposition rates on the order of 1×10^{-3} d^{-1} while the less labile soil carbon pool has decomposition rates on the order of 1×10^{-5} d^{-1} due to homogenization to low quality, highly recalcitrant carbon compounds (Webster et al., 1999; Six and Jastrow, 2002; Jackson and Vallaire, 2007; Rier et al., 2007; Yoshimura et al., 2008; Venarsky et al., 2012).

Water within the conduit studied was highly oxygenated during the study period with measurements showing levels at or near saturation much of the time, and on average dissolved oxygen was 76% its saturation level. A year-round fish population exists within the conduit, as visualized using downhole video, further supporting the oxygenated conditions. The oxygenated conditions coupled with the presence of the surficial fine grained laminae support the concept of oxidation of labile sediment carbon within the karst conduit as a process for carbon loss in the fluviokarst system. Based on the distribution of carbon pools (Table 3) and estimated aerobic decomposition rate of sediment carbon, the net residence time of sediment carbon within the karst conduit was 342 (±190) days or nearly one year. This contrasts the water transport, as the karst subsurface water has an average residence time of about one day.

The one year storage of sediment and loss of carbon within the phreatic conduit support the concept that sediment carbon turns over at high rates in the subsurface and recharges degraded organic carbon back to surface streams. The authors further support this concept because alternative explanations for the decreases in carbon density can be marginalized using our other results measured for the conduit and surface streams. The near identical particle size distributions of source sediments from tributaries in the watershed and sediments collected from the karst conduit (Fig. 7) justify the idea that the same sediments are being studied at both source and sink locations and that additional sediment sources have not been erroneously omitted. Further, $\delta^{13}C$ of inflowing source sediments and $\delta^{13}C$ of outflowing conduit sediments were not significantly different (p-value = 0.79) (Fig. 10). The lack of difference for the carbon isotope signatures suggests again that the same sediments are being studied at both source and sink locations. Heterotrophically-mediated oxidation in oxygenated waters would not be expected to produce a substantial change in $\delta^{13}C$, as past studies have shown relatively small enrichment ratios and suggest that $\delta^{13}C$ of sediment carbon pools is fairly conservative (Ford et al., 2015). In the present study, the carbon isotope change can be estimated considering isotope fractionation during the carbon turnover and net loss. Considering the Rayleigh model (Eq. (4)), isotopic enrichment of temporarily stored karst sediments would result in a conservative estimate of 0 to 0.5‰ change in the sediment carbon pool. As mentioned, data results did not reflect significant changes in $\delta^{13}C$ when comparing karst inputs (−26.6 ± 0.8‰) and outputs (−26.6 ± 0.9‰). While Rayleigh fractionation does not consider variability such as that imposed by transient fractionation (Maggi and Riley, 2009), the result highlights further evidence towards the carbon turnover in the subsurface karst. Further, the lack of isotopic change supports the suggestion of aerobic, as opposed to anaerobic, carbon mineralization due to the fact that anaerobic losses result in pronounced isotope changes for the substrate (e.g., isotopic enrichment on the order of −80‰ during methanogenesis of deposited sediment carbon, Liu et al., 2013).

The data results provide a conceptual model for the behavior of sediment carbon within phreatic karst (Fig. 1). Strong physical coupling of surface streams with subsurface karst pathways promotes the pirating of terrestrially-derived sediment carbon to the karst aquifer. The limited, yet sustained, transport carrying capacity of the conduit promotes the deposition of labile carbon to the conduit bed followed by later resuspension of the degraded sediment. Year-round flow within the conduit coupled with the subsequent deposition and resuspension of sediment provide conditions for heterotrophic bacteria to oxidize labile sediment carbon and in turn provide a mechanism for particulate carbon loss. The existence of the loosely compacted surficial fine grained laminae at the floor of the conduit within oxygenated water further supports the phreatic conduits as a biologically-active pathway that degrades sediment carbon. The subterranean biology is unique relative to surface streams because there is a lack of autochthonous growth to offset heterotrophic-respired CO_2 due to the lack of sunlight. For example, the net loss of sediment carbon for the karst conduit contrasts the surface stream in a neighboring watershed where a 50% enrichment in sediment carbon occurred due to the sequestration of humified algal (Ford and Fox, 2015). Further, karst water on average is warmer than water in surface streams in this region, i.e., mean annual temperature is 16.5 and 13.7 °C for the karst conduit and a neighboring surface stream (Ford and Fox, 2015), respectively. While the mean water temperatures are just a few degrees different, microbial growth rate increases exponentially with water temperature (White et al., 1991). For example,

Table 3
Sediment organic carbon source allocation for varying percent algae. Note: urban tributaries contribute no algal load.

Agriculture Tributaries (%)			Urban Tributaries (%)		
Algae	Litter	Soil	Algae	Litter	Soil
0.0	78.3	21.7	0.0	48.0	52.0
5.0	63.4	31.6	0.0	50.1	49.9
10.0	46.8	43.2	0.0	51.1	48.9
15.0	36.0	49.0	0.0	52.3	47.7
20.0	31.0	49.0	0.0	53.5	46.5
30.0	27.3	42.7	0.0	53.7	46.3
40.0	24.5	35.5	0.0	52.9	47.1

deposited sediment carbon experiences winter water temperatures of 9.2 °C in the surface stream relative to 15.8 °C in the phreatic karst, which more than doubles the bacteria growth rate (White et al., 1991).

3.4. Implications for carbon in fluviokarst

Given the similarity of water and sediment results in this study with other studies, it is highly conceivable that other phreatic karst systems show a similar behavior in terms of sediment carbon turnover within karst pathways. One implication of biologically-active karst pathways is that karst springheads may produce a low quality sediment carbon source to stream systems. A number of studies have reported the high sediment loads that karst springheads can discharge to surface streams (Mahler and Lynch, 1999; Drysdale et al., 2001; Herman et al., 2008; Reed et al., 2010). Due to carbon turnover within karst pathways, the springheads may provide lower quality sediment carbon than would be expected from the surrounding landscapes, which in turn will impact carbon mineralization rates controlling CO_2 outgassing from streams and freshwater ecosystem function (Butman and Raymond, 2011; Raymond et al., 2013). In this manner, the presence of karst pathways should be considered as scientists attempt to estimate organic matter stocks and transformation in streams. Further, in terms of the fluvial system, which transforms carbon en route to the ocean, the phreatic karst pathway is perceived as a discontinuity due to the increased residence time of sediment (Fig. 11). Discharge has been recognized as the primary driver of differences in organic carbon spiraling lengths in low-order Midwestern agricultural streams (Griffiths et al., 2012) highlighting the potential of karst to increase turnover as a result of limited discharge. The karst pathway would lead to higher net CO_2 respiration rates early on in the fluvial continuum resulting in more highly degraded sediment carbon delivery to the ocean. Findings from our study point towards a perhaps unforeseen discontinuity impacting carbon in the fluvial continuum due to phreatic karst pathways.

A second implication of biologically-active karst pathways is the potential for CO_2 production *via* microbial oxidation to exhibit control upon karst geochemistry within the karst environment. The presence of CO_2 is well recognized to control the rate of dissolution and hence erosion of carbonate rock during karst formation (White, 2002). The source of CO_2 is often a primary question when estimating the rate of development for karst morphology (e.g., surface water, advection of CO_2 in vadose zone, conduit surficial fine grain laminae). Some studies have found an increase of CO_2 concentration with depth in karst aquifers pointing to a potential source from oxidation of surface derived organic carbon (Baldini et al., 2006; Whitaker and Smart, 2007). Baldini et al. (2006) noted the importance in spatial variability of CO_2 with higher concentrations at the cave walls where fractures may shelter CO_2 from advection and also higher concentrations near soil accumulation sites such as collapsed cave floors. Alberic and Lepiller (1998) estimated the direct dependence of limestone dissolution on carbon oxidation to be 7–29 mg $CaCO_3$ L^{-1}. Results of this study suggest that stronger coupling of karst geochemistry with microbial activity associated with sediment carbon mineralization as a driver of the reactions and a consistent source of CO_2 from temporarily trapped labile material.

While the authors suggest the potential for biologically-active karst systems and discuss implications of this idea, we also point out limitations of this research. Features of this study are characteristic of phreatic pathways with the active input of labile sediment carbon. We speculate that a non-phreatic system where conduits are located in the vadose zone may have higher flow and sediment transport capacities hence reduced deposition. Karst development in these systems could be favored by CO_2 air flow advection or diffusion rather than microbial production (Garcia-Anton et al., 2014). Further, input of geogenic or fossil carbon would be expected to be fairly inert within the karst pathway due to their recalcitrant nature and transformation timescales that are several orders of magnitude smaller than those for sediment carbon. For such non-phreatic systems and systems with varying sediment quality, more research is needed to understand the fate of carbon in karst systems. Nevertheless, the result from this study provides a concept for consideration in future studies.

We offer a final discussion point regarding the advancement of research methods applied in this paper. The progress of karst research calls for continued instrumentation and measurements within karst aquifers and at springheads to estimate hydrologic processes (White, 2002). With this goal in mind, the present study collected numerous water, sediment, and biogeochemical measurements at karst inputs, within the aquifer and at the springheads. The difficulty with investigating processes within phreatic karst systems cannot be overstated, and often research methods that strive to perform tasks as simple as water connectivity rely on postulating assumptions and using all available data to accept or refute the assumption. To this end, the traditional dye trace methods offered important first validation of our understanding of the karst system. Although expensive, drilling 20 m directly into the karst aquifer was an advantage of this study in that we could continuously monitor the fluid velocity and sediment transport by means of sediment concentration at an intermediate phreatic section of the conduit. The internal karst monitoring station allowed us to test assumptions derived from the springheads such as the pressure head during hydrologic events. We also watched hours upon hours of downhole video, which provided further context for scientific discussion among our research group and visualization of sediment deposition. Perhaps the most innovative method of this study was the use of $\delta^{13}C$ of sediment to assist with discerning carbon source and fate processes. In this study, $\delta^{13}C$ was useful for understanding the providence of carbon input to the conduit, which is a contribution that extends recent research in this area (e.g. Fox and Papanicolaou, 2007; Fox and Martin, 2015) to the karst environment. Further, coupling $\delta^{13}C$ data with isotope fractionation estimates within the karst conduit allowed further evidence of aerobic mineralization of carbon.

4. Conclusions

Hydrological processes highlighted in this paper include the following:

1) The sediment transport carrying capacity of the phreatic karst water is orders of magnitude less than surface streams during storm-activated periods. The relatively buffered fluid energy promotes pronounced deposition of fine sediments to the subsurface phreatic karst.
2) The sediment transport carrying capacity is sustained long after storm events have ceased. The result diverges the phreatic karst from surface streams and non-phreatic karst where the energy of the fluid is a power function of the flow depth.
3) The surficial fine grained laminae occurs in the subsurface karst system, much like surface streams, and includes deposition of a fine sediment layer coating the cave floor. Unlike surface streams, the light-limited conditions of the subsurface karst promote constant heterotrophy leading to net degradation of sediment organic carbon.

Results of this study help provide a conceptual model for sediment carbon fate in phreatic karst. Karst pathways act as biologi-

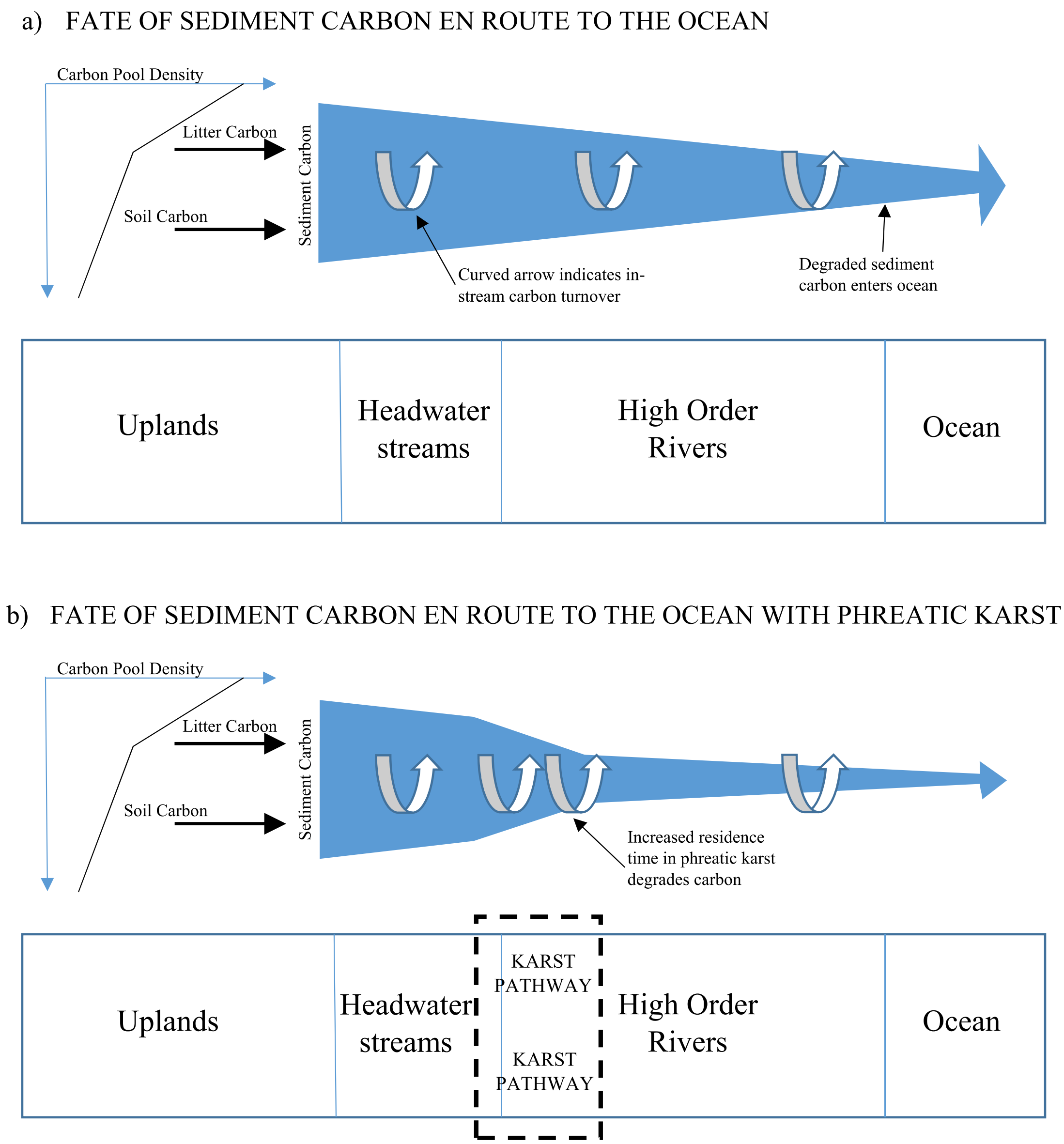


Fig. 11. Schematic of the fate of sediment organic carbon in the fluvial environment with and without the phreatic karst. (After Fox and Ford, 2016; Marín-Spiotta et al., 2014).

cally active conveyors of sediment carbon that temporarily stores sediment, turns over carbon at higher rates than would be considered otherwise, respires carbon dioxide to the water column, and recharges degraded organic carbon back to surface streams. Karst morphologic and hydrologic features for which this conceptual model are deemed applicable include hydrologic connectivity of surface streams to subsurface karst such that an active sediment delivery system exists, the presence of a phreatic system that promotes storage and turnover of carbon, and active recharge of sediment back to the fluvial network. In the case of the new data results presented here, the conceptual model is supported by a one-day residence estimated for water within the subsurface karst but nearly a one year residence estimated for sediments. Further, fluvial sediment is estimated to lose 30% of its carbon by mass during temporary residence within the subsurface karst prior to recharge back to the surface streams.

Implications of the conceptual model proposed here are that karst springheads produce low quality sediment carbon source to stream systems, which in turn impacts carbon mineralization rates controlling CO_2 outgassing from streams and freshwater ecosystem function. A second implication of biologically-active karst pathways is the potential for CO_2 production *via* microbial oxidation to exhibit control upon karst geochemistry within the karst environment. It is recommended that scientists consider karst pathways when estimating carbon and carbonate transformation in surface and karst streams.

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